Outlook

RE: Biopharmaceutical Proc Dev & Mfg- New course

From Nere, Nandkishor K <nandkishor.k.nere@abbvie.com>
Date Wed 3/5/2025 2:26 PM
To Zdunek, Alan <zdunek@uic.edu>
Cc Singh, Meenesh R <mrsingh@uic.edu>

1 attachment (19 KB)
UIC-Sping-2024-CHE 494-Course-contents.docx;

CAUTION: External Sender

Hi Alan,
Thank you for all your help. Here follows the information in support of the CRS.

1. **Course learning outcome assessment methods**:
    a. Participation in class discussions
    b. Team projects
    c. Project presentations
    d. Learnings from the mandated interactions with presenting teams
2. **Course Materials**:
    a. No required textbooks
    b. Course material will be provided in the form of PDF slides
    c. Reference Books (optional):
        i. D. J. am Ende and M. T. am Ende (Eds.), Chemical Engineering in the Pharmaceutical Industry: Active Pharmaceutical Ingredients, 2nd Edition, Hoboken, NJ, Wiley.
        ii. AAPS Advances in the Pharmaceutical Sciences Series: Formulation and Process Development Strategies for Manufacturing Biopharmaceuticals, Jameel, F., Hershenson, S. (Eds.) 2010, Springer Nature Switzerland AG.
        iii. D. J. am Ende and M. T. am Ende (Eds.), Chemical Engineering in the Pharmaceutical Industry: Drug Product Design, Developmentand Modeling, 2nd Edition, Hoboken, NJ, Wiley.
        iv. AAPS Advances in the Pharmaceutical Sciences Series: Quality by Design for Biopharmaceutical Drug Product Development, Jameel, F.,Hershenson, S.,Khan, M. A.,Martin-Moe, S. (Eds.) 2015, Springer Nature Switzerland AG.
        v. AAPS Advances in the Pharmaceutical Sciences Series: Development of Biopharmaceutical Drug-Device Products, Jameel, F., Skoug, J. Nesbitt, R. (Eds.) 2020, Springer Nature Switzerland AG.
3. **Course Materials**:
    a. Introduction to biopharmaceuticals process development- 3 hours
    b. Introduction to biopharmaceuticals manufacturing and cGMP considerations- 3 hours
    c. Introduction to Process Analytical Technologies -3 hours

d. Active pharmaceutical Ingredient (API) Crystal Form Screening and Selection -3 hours
e. Small Molecule Drug substance (active pharmaceutical ingredients) process development- 12 hours
f. Formulation development for oral solid dosage forms -3 hours
g. Small Molecule Drug product process development -3 hours
h. Biologics upstream and downstream process development -3 hours
i. Enabling technologies and advances in predictive modeling for bioprocess development and manufacturing -6 hours
j. Biopharmaceutical process scale-up covering drug substance and drug product -3 hours
k. Exam- Project presentation and Interactions with other project teams- 3 hours

Thank you very much for the draft CRS. Some suggestions/changes (tried to highlight in blue as a ready reference) for your further consideration:

- **Cross listings: Should we add other departments (Pharm Sci, Biomedical/Biochemical Eng, Material Science, etc.) in here as well?**
- **Type of Course**: Selective for the following programs: Selective in the ~~Nanotechnology~~ Biotechnology/Bioprocess/Biochemical engineering/Pharmaceutical Science concentration and a technical elective in the BS chemical engineering degree course
- **Learning Outcomes:** 1. Students will be able to evaluate active pharmaceutical ingredients using extensive solid form screening and various characterization techniques 2. Students will be able to apply process development methods to drug development and scale-up 3. Students will be able to understand current Good Manufacturing Practice (cGMP) and identify safety considerations. 4. Students will be able to embrace team work to solve problems in multidisciplinary setting.
- **Purpose of this course in relation to overall curriculum and relationship to other courses offered by primary unit:** This course serves as both a selective course for the Biotechnology/Bioprocess/Biochemical engineering/Pharmaceutical Science ~~Nanotechnology~~ concentration and a technical elective in the core BS chemical engineering degree. The course is designed to provide fundamental knowledge of process development and manufacturing in the biopharmaceutical industry.
- **Restrictions:** Restricted to students in the following colleges/schools: Engineering, Pharmaceutical Science, Materials Science or Graduate College.

Also attached is the syllabus if that helps. Please let me know if you have any questions or need clarification.

Thanks and regards,
Nandu
R&D Process Engineering Communication Channel

I work across a number of time zones, so I am sending this message now because it works for me. Feel free to read, act on, or respond during your working hours.

"It's amazing what you can accomplish if you do not care who gets the credit." - Harry Truman



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From: Zdunek, Alan <zdunek@uic.edu>
Sent: Wednesday, March 5, 2025 9:10 AM
To: Singh, Meenesh R <mrsingh@uic.edu>; Nere, Nandkishor K <nandkishor.k.nere@abbvie.com>
Subject: [EXTERNAL] Re: Biopharmaceutical Proc Dev & Mfg- New course

Hello Meenesh,

Just following up on my email about creating a permanent course for Biopharm Proc Dev & Mfg, I am cc'ing Nandkishor as well.

I filled in several of the sections in the attached course request document; please review to make sure I am capturing the details of the course correctly.

Also, I need information on the **course learning outcome methods** (exams, projects, etc) and the **course material** (textbook, notes, etc).
If you have a syllabus to share that would be helpful.

There is an Education Policy Committee (EPC) meeting next week Monday (3/10), so if you can give me input before then I can put it on the agenda for approval.

Thanks,

Alan

Dr. Alan D Zdunek
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From: Zdunek, Alan
Sent: Monday, March 3, 2025 5:51 PM
To: Singh, Meenesh R <mrsingh@uic.edu>
Subject: Biopharmaceutical Proc Dev & Mfg- New course

Hi Meenesh,

I am working on the CRS (course request system) form so that we can make your CHE 494 into a permanent course, and I need input from you and Nandkishor.

Specifically, please review and add/edit these sections in the form:

Course description (limited to 25 words)
Course Learning Outcomes (Format is "Students will be able to ...action verb"
Course Learning Outcome Methods: Mapped to Course Learning Outcomes
Major Topics : Must add up to 45 hours
Sample sources/materials/textbooks, etc that you use.

If you have any questions, let me know.

Thanks,

Alan

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